EARTHPRESS · RISK REGISTER AND HAZARD ANALYSIS

EarthPress risk register and hazard analysis

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-30	Amish Chadha	R3 rescored after decision D2 (205 kg total, 2026-10-30): margin 1.2 kg
0.1	2026-10-25	Amish Chadha	First risk register, preliminary hazard analysis and first-pass design FMEA (kit 1.8 pipeline, stage P3) for option A after fix F1

EarthPress risk register and hazard analysis

The top project risks are that partner soils turn out softer or stiffer than assumed, which moves the operating force and the fill window (R1), and that the press stays close to its mass limit and over its cost target (R3, R4). The top hazards are the stored energy at the end of the stroke (lever kickback, about 47 J), crushing in the linkage and the mold, and silica dust and wet cement during mixing. The design FMEA's highest risk priority numbers are the fill weighed outside its window and a worn or missing latch pin. Method: STANDARDS section 23 and the ISO 12100 approach. Likelihood, impact, severity and occurrence are scored 1 (low) to 5 (high); detection 1 (certain to be detected) to 5 (not detected).

Project risk register

Table 1. Project risks.

#	Risk	Cause	Effect	Likelihood	Impact	Owner (role)	Response
R1	Partner soils differ from the assumed compaction law	The law (tenfold over the last 14 mm) is assumed (STM S6)	Peak pull above 500 N per operator or a narrower fill window	3	4	Design lead with the co-design partner	Measure the curve on partner soils first (TRL 4, on hold); the operating chart already uses the softer soil
R2	No co-design partner engaged	Partner not yet approached	Requirements stay unvalidated; no soils or tests	3	4	Amish	Approach the first candidate (decided 2026-10-02) when TRL 4 opens
R3	Mass over the R7 total	Thin margin: bushes, fasteners and weld metal counted; fix F1	203.8 kg against 205 kg (D2 decided 2026-10-30)	3	2	Design lead	Weigh the prototype; lighten as a value-engineering item if it grows
R4	Cost over the value-engineering target	Steel mass, turned alloy pins, bushes	USD 542 against USD 500	4	2	Design lead	Local quotes; value-engineering items in EPR-DEC-001
R5	Concept choice reopened	Options A, B and C within 10 % (D1)	Redesign if B or C is chosen	2	4	Amish (decision D1)	Decide D1; the repo stays on A meanwhile
R6	Local parts not available	42CrMo4 bar, 40 x 46 mm bushes or UPN 80 not stocked	Substitutions change the checks	3	3	Fabricator	Confirm when parts are bought (EPR-DEC-001 Table 2); recheck with `sizing.py`
R7	Blocks below the strength reference	Unsuitable soil, low cement, poor curing	Weak walls if blocks are used untested	3	5	Co-design partner, project engineer	Field tests, mix chart, block tests before structural use (PR-003); stated in every safety note
R8	Output below 300 a day	Cycle longer than the assumed 85 s	Slower house building	3	2	Co-design partner	Timed cycles at TRL 4 (on hold)

Preliminary hazard analysis

Table 2. Hazards (option A).

#	Hazard	Cause	Harm	Severity	Likelihood	Control	Residual
H1	Lever kickback	Lever released at the end of the stroke; 47 J returned from soil rebound and column stretch	Struck head or arm	4	3	End pawl drops under the crank pin at the stop; rest catch at the start; rule: engage the pawl before letting go; heads out of the lever arc	4 x 1
H2	Crushing in the linkage	Hands in the crank, connecting link and knee scissor while the lever moves	Crushed fingers or hand	4	3	Linkage guard with openings of 12 mm or less; only the people on the lever touch it, after calling "clear"	4 x 1
H3	Lid thrown open under load	Latch unpinned while the toggle is loaded; worn or missing latch pin	Struck by the 27 kg lid or ejected soil	5	2	Lid and latch rated 173 kN; rule: rest catch in before the latch is opened; latch pin on a T-handle, inspected	5 x 1
H4	Crushing in the mold	Hands in the mold during pressing or ejection	Crushed fingers	4	2	Fill with a scoop; lid closed before pressing; eject only with the lever in the eject socket	4 x 1
H5	Eject arm and push rod	Hands near the push rod foot during ejection	Pinched fingers	3	2	Eject seesaw under the base; hands on the lever only	3 x 1
H6	Silica dust	Sieving dry soil	Lung disease over time	4	3	Sieve damp soil; work upwind; FFP2 or N95 respirator (STM S19)	4 x 2
H7	Wet cement	Mixing cement-stabilized soil by hand	Skin burns, eye damage	3	3	Gloves, long sleeves, eye protection (STM S20)	3 x 1
H8	Manual handling	Moving press pieces of up to 44.5 kg; carrying 7.6 kg blocks all day	Back injury	3	3	Two people to each piece (STM S21); rotate tasks; lift blocks close to the body	3 x 2
H9	Structural overload	Extra people or a cheater bar on the lever beyond 1,000 N	Pin, link or lid failure	5	1	Crank stop caps the force at the design load; members sized at 173 kN with factors of 1.29 or more (PR-016, fix F1)	5 x 1
H10	Fabrication	Welding, cutting, grinding	Burns, eye injury, fire	4	2	Welding PPE, guarded grinders, no flammables (BLD-001)	4 x 1
H11	Unsafe walls	Blocks used structurally without tests or design	Wall collapse	5	2	Every document states that blocks are not certified and need tests and an engineer for structural use	5 x 1

Design FMEA (first pass, P3)

Table 3. Failure modes. RPN = severity x occurrence x detection.

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
F1	Toggle pins	Carry the link forces	Bend or yield	Linkage jams; block height lost	Pins too small in bending (found in the STEM check)	5	1	3	15	Done: 40 mm 42CrMo4 +QT pins (fix F1); check straightness at each service
F2	Bushes	Pin bearing	Wear or scoring	Play in the linkage; low pressure at the stop	Grit, lack of grease	3	3	3	27	Dust seals, grease nipples, weekly greasing; measure play at TRL 4
F3	Fill	Set block density and height	Fill outside 0 to +2.7 %	Weak block (under) or operators stall short of the stop (over)	Not weighed; scale misread	4	4	3	48	Weigh every fill; scoop only after the scatter test; block gauge on every block
F4	Latch pin	Hold the lid	Worn, bent or missing	Lid thrown open (H3)	Wear, loss	5	2	4	40	Inspect at each shift; spare pin on a chain; replace when worn by 0.5 mm
F5	End pawl and spring	Hold the crank pin at the stop	Spring broken; pawl does not drop	Kickback (H1)	Fatigue, grit	4	2	3	24	Check pawl action at the start of each shift; spare spring
F6	Mold walls	Shape the block	Wear of the cavity faces	Oversize blocks	Sandy soils	3	3	2	18	Gauge blocks; measure wear against PR-008 (decided 2026-10-02)
F7	Welds of the base beam and lugs	Carry the toggle reaction	Fatigue crack	Loss of the base pin support	Poor weld; overload	5	1	4	20	FAT 71 detail at 89 MPa range (S11); visual check of the lug welds each month
F8	Mold bolts	Hold the mold to the columns	Loosen	Mold shifts; flange wear	Vibration, no locking	3	2	3	18	Torque to grade 8.8 value; check monthly
F9	Lever socket and locking pin	Hold the lever	Pin missing	Lever slides out under load	Pin lost	4	2	2	16	Clevis pin with split pin; spare on a chain
F10	Guard	Keep hands out	Removed or damaged	Crushing (H2)	Removed for cleaning and not refitted	4	2	2	16	Rule: no pressing without the guard

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		of the linkage								